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PRS821 PROGRAMMING FOR SECURITY PERSONNEL, SEM 1, 2026 NETWORK ATTACKS (WEEK 5D)

WHAT IS NETWORK ATTACKS?

Network attacks are unauthorized actions on the digital assets within an organizational network. Malicious parties usually execute network attacks to alter, destroy, or steal private data. Perpetrators in network attacks tend to target network perimeters to gain access to internal systems.

DIFFERENT TYPES OF NETWORK ATTACKS

Modern organizations rely on the internet for communication, and confidential data is often exchanged between networks. Remote accessibility also provides malicious parties with vulnerable targets for data interception. These may violate user privacy settings and compromise devices connected to the internet.

Network attacks occur in various forms. Enterprises need to ensure that they maintain the highest cybersecurity standards, network security policies, and staff training to safeguard their assets against increasingly sophisticated cyber threats.

Your Data may be compromised without security measures and controls. In other words, the information is active so that the information is altered to corrupt or destroy the data or the network itself. The data is,

therefore, actively monitored. For many of their jobs, social and personal activities, numerous people rely on the Internet.

Some people are also trying to harm our computers connected to the Internet, breach our privacy and make internet services inoperative. In the field of computer networking, network security has become a key issue in the intensity and scope of current attacks and the risk of new and more damaging future attacks. If you have no security plan, the networks and Data are vulnerable to any of the following attacks.

Basic Network Attacks in Computer Network

Many people rely on the Internet for many of their professional, social and personal activities. But there are also people who attempt to damage our Internet-connected computers, violate our privacy and render inoperable the Internet services.

Given the frequency and variety of existing attacks as well as the threat of new and more destructive future attacks, network security has become a central topic in the field of computer networking.

How are computer networks vulnerable? What are some of the more prevalent types of attacks today?

Virus – A malware which requires some form of user's interaction to infect the user's device. The classic example is an e-mail attachment containing malicious executable code. If a user receives and opens such an attachment, the user inadvertently runs the malware on the device.

Computer Virus Examples

Imagine your computer as a planet in the vast universe of the internet. Now, consider a comet – a computer virus – hurtling towards it. On impact, it can reshape the planet or even wipe out all life (data and programs). This might sound like a sci-fi movie, but such scenarios have happened in the real world.

In this article, we'll recount the tales of the most devastating comets – the most notorious computer viruses – that have ever struck the digital planets.

Summary

A computer virus is a piece of malware that infects files and programs on your computer. Opening an infected file will execute the virus code and result in damage to your files, your computer, and your overall internet security.

- a. **Elk Cloner** (1982): The first known microcomputer virus that spread “in the wild,” or outside the computer system or lab in which it was written.*
- b. **Brain** (1986): The first MS-DOS based virus, which targeted IBM PC systems.*
- c. **Morris Worm** (1988): One of the first worms distributed via the Internet, it was created by Robert Tappan Morris.*
- d. **CIH or Chernobyl Virus** (1998): A destructive virus that rendered machines unbootable.*
- e. **ILOVEYOU or Love Letter Virus** (2000): A worm that attacked tens of millions of Windows PCs via email.*
- f. **Code Red and Code Red II** (2001): Worms that exploited a vulnerability in Microsoft's Internet Information Server (IIS).*
- g. **Slammer or Sapphire** (2003): A worm that caused a denial of service on some Internet hosts and dramatically slowed down general Internet traffic.*
- h. **Blaster Worm or MSBlast** (2003): A worm that spread via a vulnerability in Windows systems, causing a significant amount of network traffic.*
- i. **Sobig.F** (2003): A worm and trojan that circulated through emails as viral spam, it was one of the most widely circulated worms.*
- j. **Mydoom** (2004): An extremely rapidly spreading email-based worm.*

- k. **Sasser and Netsky** (2004): A series of worms that caused problems in networks by exploiting a vulnerability in Windows systems.
- l. **Conficker** (2008): A worm that targeted Windows and caused a range of problems from consuming network resources to downloading malware.
- m. **Stuxnet** (2010): A highly sophisticated worm that targeted industrial control systems, it was responsible for causing substantial damage to Iran's nuclear program.
- n. **CryptoLocker** (2013): A ransomware that encrypted users' files and then demanded a ransom to unlock them.
- o. **WannaCry** (2017): A ransomware worm that spread rapidly across numerous networks, infecting hundreds of thousands of computers worldwide.
- p. **NotPetya** (2017): Masqueraded as ransomware but was designed to cause damage, primarily targeted Ukraine but had global effects.
- q. **Bad Rabbit** (2017): A ransomware attack, believed to be a variant of NotPetya, that spread through "drive-by downloads" where insecure websites were targeted.

Don't become a victim of a computer virus. Protect your devices with the best antivirus software and your privacy with the best VPN service.

Computer Virus vs Malware

The terms "computer virus" and "malware" are often used interchangeably, but they are not the same. A virus is a type of malware, but not all malware are viruses.

Malware is a general term for any malicious software, including viruses, worms, trojans, ransomware, adware, and spyware. A computer virus, specifically, is a type of malware that can copy itself and infect a computer without the permission or knowledge of the user.

In the list we provided earlier, some examples like Stuxnet, Conficker, and Slammer are technically worms, not viruses. Worms, unlike viruses, don't

require user action to propagate; they can self-replicate and spread independently.

Similarly, CryptoLocker, WannaCry, NotPetya, and Bad Rabbit are examples of ransomware, another type of malware that encrypts users' files and demands a ransom for their release.

Malware – short for malicious software which is specifically designed to disrupt, damage, or gain authorized access to a computer system. Much of the malware out there today is self-replicating: once it infects one host, from that host it seeks entry into other hosts over the Internet, and from the newly infected hosts, it seeks entry into yet more hosts. In this manner, self-replicating malware can spread exponentially fast. Some examples of Malware are as follows:

Zeus Malware:

Zeus, otherwise called Zbot, is a malware bundle that utilizes a client/server model. Programmers utilize the Zeus malware to make gigantic botnets. The primary reason for Zeus is to assist programmers with acquiring unapproved admittance to monetary frameworks by taking accreditations, banking data and monetary information. The penetrated information is then sent back to the assailants through the Zeus Order and Control (C&C) server.

Zeus has tainted north of 3 million PCs in the USA, and has compromised significant associations like NASA and the Bank of America.

Cobalt Strike: White Cap Programmer Force to be reckoned with in Some unacceptable Hands:

Cobalt Strike is a business infiltration testing instrument. This instrument empowers security analyzers admittance to a huge assortment of assault capacities. You can utilize Cobalt Strike to execute stick phishing and gain unapproved admittance to frameworks. It can likewise recreate an assortment of malware and other high level danger strategies.

While Cobalt Strike is a real instrument utilized by moral programmers, some digital hoodlums get the preliminary rendition and break its product insurance, or even get admittance to a business duplicate of the product.

FTCode Ransomware: Dispersion, Life systems and Assurance

FTCode is a kind of ransomware, intended to encode information and power casualties to pay a payoff for a decoding key. The code is written in PowerShell, implying that it can scramble records on a Windows gadget without downloading some other parts. FTCode loads its executable code just into memory, without saving it to plate, to forestall location by antivirus. The FTCode ransomware is conveyed through spam messages containing a contaminated Word layout in Italian.

Mimikatz is an open-source instrument at first created by moral programmer Benjamin Delpy, to exhibit a blemish in Microsoft's confirmation conventions. .As such, the apparatus takes passwords. It is conveyed on Windows and empowers clients to extricate Kerberos tickets and other validation tokens from the machine. A portion of the more significant assaults worked with by Mimikatz incorporate Pass-the-Hash, Kerberos Brilliant Ticket, Pass the Key, and Pass-the-Ticket.

Worm – A malware which can enter a device without any explicit user interaction. For example, a user may be running a vulnerable network application to which an attacker can send malware. In some cases, without any user intervention, the application may accept the malware from the Internet and run it, creating a worm.

Phishing – The fraudulent practice of sending emails purporting to be from reputable companies in order to induce individuals to reveal personal information, such as passwords and credit card numbers.

Botnet – A network of private computers infected with malicious software and controlled as a group without the owners' knowledge, e.g. to send spam.

The advance types of network attacks are as follows:

1. Password-Based Attacks

Password-based access control is the common denominator of most network and operating system **security policies**. You can, therefore, determine who you are, that is, your user name and your password, your computer and your network access rights. Old systems do not always secure identity information because authentication information is transmitted through the network.

This could give an eavesdropper legitimate user access to the network. The intruder has the same privileges as an actual client if he enters a legitimate user account. Therefore, the intruder may also build later access accounts if the client has administrator-leaved privileges.

An attacker can do any of the following after accessing your network with a legitimate account.

Use lists of the relevant client and network data and device names.
Modify database and network configuration, including routing and access controls.

Adjust your info, re-route it, or delete it.

2. Man-in-the-Middle Attack

As its name suggests, when someone in the center is constantly tracking, capturing and monitors your contact, someone between you and the person with whom you interact. The attacker can, for instance, restart the data exchange. Computers can not determine how they share

information on a low level of the network layer While computers are communicating.

Man-in – the- middle attacks are just like those who take up your identity to read your text. The other person might assume you are because the intruder may deliberately respond so that you sustain the exchange and obtain more information. His attack can do the same damage as an app layer attack mentioned below in this section.

Man-in-the-middle (MITM) network attacks occur when malicious parties intercept traffic conveyed between networks and external data sources or within a network. In most cases, hackers achieve man-in-the-middle attacks via weak security protocols. These enable hackers to convey themselves as a relay or proxy account and manipulate data in real-time transactions.

3. Close-in Attack

A Close-in Attack involves someone who attempts to physically enter the elements, data or structures of a network to find out more about a close-in attack consists of ordinary persons entering near physical proximity to networks, systems or facilities to alter or collect information or to reject access. Near physical proximity is achieved by sudden network entry, open access, or both. A popular form of close attack is social engineering in a social engineering attack. Through social interaction, an email message or a telephone, the attacker exploits the network and device.

A Close-in Attack means someone who attempts to enter devices, data, or systems in a physical manner to learn about the Close-in Attack consists of individuals who are frequently close to networks, systems, or installations to alter, capture or deny access to information. Near physical closeness is reached by surreptitious open access, network access or both.

A Social engineering attack is a common form of attack, where the aggressor jeopardizes the network or process through social interaction with an individual, e-mail or mobile. The person can use various tricks to expose company security information. The information the victim provides to the attacker is most likely to be used to access an unauthorized device or network for subsequent attacks.

4. Identity Spoofing

The IP address of a device is used to classify a legitimate business by most networks and operating systems. An intruder can also create IP packets from valid addresses in the corporate intranet using specific programs. An attacker can. The hacker may alter, remove, or erase your data after accessing the network using a valid IP address. As defined in the following sections, the attacker may also perform other Types of Attacks.

5. Compromised-Key Attack

One key is a secret code or number required for the processing of secure information. While it is possible to obtain a key for an attacker to be a complicated and resource-intensive process. After an attacker gets a password, it is considered a corrupted key.

An attacker uses the affected key to gain access to secure communication without the attack being detected by the sender or recipient. The attacker may decrypt or alter the information by using the affected key to generate additional keys to give the attacker access to any other secure communications.

6. Application-Layer Attack

An application-layer attack targets database servers, triggering a failure on a server's operating system or applications deliberately. It helps the intruder to bypass standard access controls. This situation is used by the intruder, who gets control of your application, device or network and can do any of the following:

- Read your data or add, operating system, delete or change them.
- Introduce a virus system to copy viruses in your network using your computers and software applications.
- Introduce a sniffer to evaluate and collect information that can crash or corrupt the network and systems in the end.

7. *Passive Attack*

A Passive Attack tracks unencrypted traffic and scans for code or confidential information for other attack forms. Passive threats include traffic analyzes, insecure contact surveillance, weakly encrypted traffic decryption, and encryption information collecting, for example, passwords.

Passive network monitoring allows opponents to see future measures. Passive attacks lead, with no user consent or knowledge, to the disclosure of information or data files to an attacker.

8. *Active Attack*

The Attacker attempts to hack or crack into secure systems in an aggressive attack. It can take place through stealth, worms, viruses or Trojan horses. Aggressive attacks include attempts to circumvent or break safety software, malicious codes, and theft or alteration. Such attacks have been installed on a network backbone, take advantage of the information in transit, join an enclave electronically or target a remote authorized user while attempting to link to an enclave. Active attacks lead to data files, DoS and alteration. Software is exposed and disseminated.

9. *DoS (Denial of Service)*

A **DoS** Attack renders legitimate users unable to use a network, server or other resources. In one of the three groups.

- a) **Bandwidth Flooding:** The Attacker sends a deluge of packets to the target host — so many packets that the access path to the target is blocked, and legit packets can not enter the server.
- b) **Vulnerability Attack:** This means sending a set of well-constructed messages on the targeted host to a vulnerable program or operating system. If a compromised program or operating system is sent the correct sequence of packages, the service can stop, or the host can crash.
- c) **Connection Flooding:** Many TCP connections on the target host are formed half-open or completely open. With these fake connections, the host can be so enmeshed that it can no longer accept valid connections.

10. DDoS (Distributed Denial of Service) attacks involve deploying sprawling networks of botnets — malware-compromised devices linked to the internet. These bombard and overwhelm enterprise servers with high volumes of fraudulent traffic. Malicious attackers may target time-sensitive data, such as that belonging to healthcare institutions, interrupting access to vital patient database records.

11. Packet Sniffer

A passive receiver that records a copy of each flying packet is a packet sniffer. By every passive receiver near the wireless transmitter, it can get a copy of each transmitted packet. Such packages can contain some sensitive information such as social security numbers, passwords, personal messages, and business secrets. Cryptography includes some of the best defences from packet sniffing.

12. Malware

Malware is specifically intended for interrupting, damaging or obtaining licensed computer system access. Some of the malware today replicates itself: Once the host becomes infected, it is looking for connections to other hosts via the internet from that host and seeks entry in even

more **hosts** from the newly infected host. Self-replicating malware can propagate exponentially rapidly in this way.

13. Insider Attack

Insider Attacks involve someone from inside of the company or system, such as an insecure worker who may be malicious or not malicious by targeting the network for insider attacks. Intentional malicious insiders eavesdrop, steal data or erase it, fraudulently use it or deny access to other users who have been licensed. There are no traditional malicious attacks due to lack of consideration, awareness or intentional security circumvention, for example, executing a mission.

14. Unauthorized Access

Unauthorized access refers to network attacks where malicious parties gain access to enterprise assets without seeking permission. Such incidences may occur due to weak account password protection, unencrypted networks, insider threats that abuse role privileges, and the exploitation of inactive roles with administrator rights.

Organizations should prioritize and maintain the least privilege principle to avoid the risks of privilege escalation and unauthorized access.

15. SQL Injection

Unmoderated user data inputs could place organizational networks at risk of SQL injection attacks. Under the network attack method, external parties manipulate forms by submitting malicious codes in place of expected data values. They compromise the network and access sensitive data such as user passwords.

There are various SQL injection types, such as examining databases to retrieve details on their version and structure and subverting logic on the application layer, disrupting its logic sequences and function.

Network users can reduce the risks of SQL injection attacks by implementing parameterized queries/prepared statements, which helps

verify untrusted data inputs.

16. IP Spoofing – The ability to inject packets into the Internet with a false source address is known as IP spoofing, and is but one of many ways in which one user can masquerade as another user. To solve this problem, we will need end-point authentication, that is, a mechanism that will allow us to determine with certainty if a message originates from where we think it does.

17. DNS spoofing – Also referred to as DNS cache poisoning, is a form of computer security hacking in which corrupt Domain Name System data is introduced into the DNS resolver's cache, causing the name server to return an incorrect IP address.

18. Compromised-Key Attack – A key is a secret code or number necessary to interpret secured information. Although obtaining a key is a difficult and resource-intensive process for an attacker, it is possible.

After an attacker obtains a key, that key is referred to as a compromised key.

An attacker uses the compromised key to gain access to a secured communication without the sender or receiver being aware of the attack.

19. Rootkit – Rootkits are stealthy packages designed to benefit administrative rights and get the right of entry to a community tool. Once installed, hackers have complete and unrestricted get right of entry to the tool and can, therefore, execute any movement including spying on customers or stealing exclusive data with no hindrance.

20. Command injection is a cyber attack that involves executing arbitrary commands on a host operating system (OS). Typically, the

threat actor injects the commands by exploiting an application vulnerability, such as insufficient input validation.

Summary

It's a vision to think a network infrastructure is invulnerable, but it's within reach of the possibility of safety. Essentially, you know what your network can do, know your equipment and train your workers. So, in this article, we have seen what different types of Network Attacks are. I hope you will find this article helpful.

RECENT NETWORK ATTACKS

Network attacks remain a lingering issue for organizations as they transition to remote operations with increased reliance on confidential network communications. Recent network attacks demonstrate that malicious parties may strike at the least expected moment. So, cyber vigilance and security should be a priority across all industries.

Social Engineering

According to ISACA's State of Cybersecurity 2020 Report, social engineering is the **most popular network** attack method, with 15 percent of compromised parties reporting the technique as the vehicle of infiltration. **Social engineering** involves elaborate techniques in deception and trickery techniques — phishing — that leverage users' trust and emotions to gain access to their private data.

Advanced Persistent Threats

Some network attacks may involve **advanced persistent threats** (APTs) from a team of expert hackers. APT parties will prepare and deploy a complex cyber-attacks program. This exploits multiple network vulnerabilities while remaining undetected by network security measures such as firewalls and antivirus software.

Ransomware

In **ransomware** attacks, malicious parties encrypt data access channels while withholding decryption keys, a model that enables hackers to extort affected organizations. Payment channels usually include untraceable cryptocurrency accounts. While cybersecurity authorities discourage paying off malicious parties, some organizations continue to do so as a quick solution in regaining data access.

PROTECTION FROM NETWORK ATTACKS

Evolving network attacks require a modern and proactive **network security** solution. **Forcepoint's NGFW** (Next Generation Firewall) provides modern organizations with a suite of sophisticated features necessary to detect and respond to the most insidious threats within a network.

The NGFW's real-time monitoring interface enables users to react quickly to the slightest network anomalies without delay, with a clear breakdown of ongoing processes. NGFW prioritizes critical networks and devices while identifying the most evasive network attacks that bypass conventional firewalls.

Additionally, Forcepoint's next-gen firewall solution safeguards user privacy while operating decryption functions that effectively spot potentially stolen or compromised data within SSL and TLS traffic.

Avoid camouflaged network attacks with a firewall solution built to close the evasion gap. Experience the Forcepoint method to optimize your enterprise data security standards through its digital transformation.

Since cyber attackers are now using Artificial Intelligence (AI) tools to carryout attacks, it is essential for organisations to upgrade their protection mechanism from conventional defense to AI-based defense mechanism.

